LAB EXPERIMENTS

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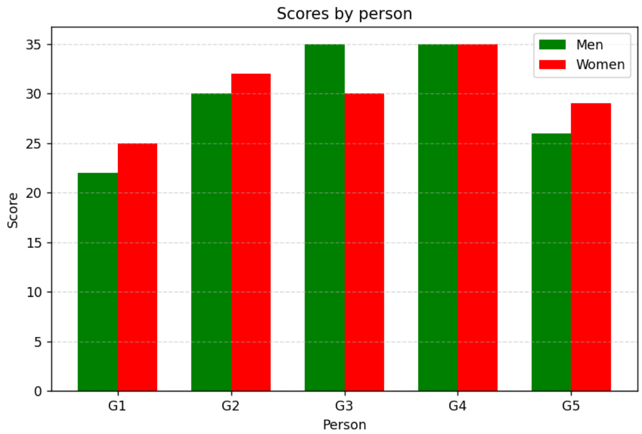
30. Write a Python program to create bar plot of scores by group and gender. Use multiple X values on the same chart for men and women.

Code :

```
import numpy as np
import matplotlib.pyplot as plt

# Sample data
men = (22, 30, 35, 35, 26)
women = (25, 32, 30, 35, 29)

# X-axis positions
x = np.arange(5)
width = 0.35

# Create bar chart
plt.figure(figsize=(8, 5))

plt.bar(x - width/2, men, width, label='Men', color='green')
plt.bar(x + width/2, women, width, label='Women', color='red')
```

```
# Labels and title
plt.xlabel('Person')
plt.ylabel('Score')
plt.title('Scores by person')

# X-axis labels
plt.xticks(x, ['G1', 'G2', 'G3', 'G4', 'G5'])

# Grid
plt.grid(axis='y', linestyle='--', alpha=0.5)

# Legend
plt.legend()

# Display chart
plt.show()
```

OUTPUT :



31.Write a Python program to create a stacked bar plot with error bars.

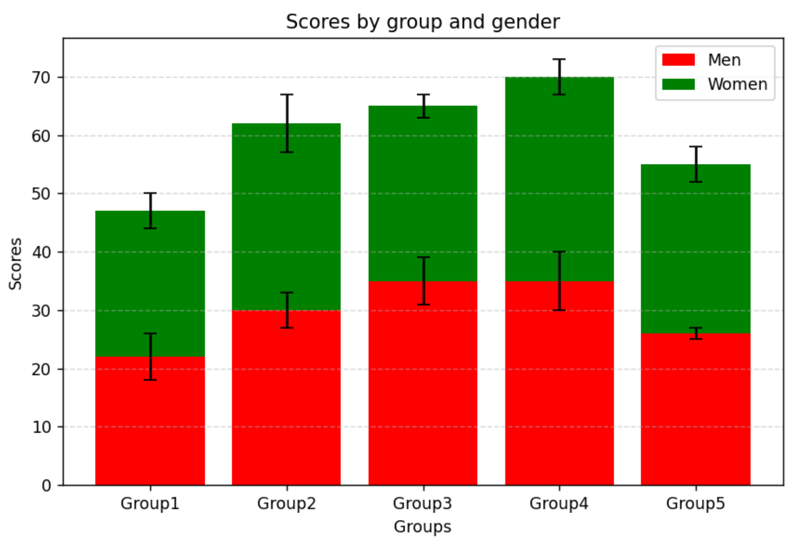
Note: Use bottom to stack the women? bars on top of the men? bars.

Code :

```
import numpy as np
import matplotlib.pyplot as plt

# Sample data
men = (22, 30, 35, 35, 26)
women = (25, 32, 30, 35, 29)
# Standard deviations
men_std = (4, 3, 4, 5, 1)
women_std = (3, 5, 2, 3, 3)
# X-axis positions
x = np.arange(5)
# Create stacked bar chart
plt.figure(figsize=(8, 5))
# Men's bars
plt.bar(
    x,
    men,
    yerr=men_std,
    capsize=4,
    label='Men',
    color='red')
# Women's bars stacked on top of men's bars
plt.bar(
    x,
```

```
    women,
    bottom=men,
    yerr=women_std,
    capsize=4,
    label='Women',
    color='green')
# Labels and title
plt.xlabel('Groups')
plt.ylabel('Scores')
plt.title('Scores by group and gender')
# X-axis labels
plt.xticks(x, ['Group1', 'Group2', 'Group3', 'Group4', 'Group5'])
# Grid
plt.grid(axis='y', linestyle='--', alpha=0.5)
# Legend
plt.legend()
# Display graph
plt.show()
```

OUTPUT :



32) Python Program – Bar Chart with Different Colors

Code :

```
import numpy as np

import matplotlib.pyplot as plt

# Generate random data for X and Y

np.random.seed(10)

x = np.random.randn(200)

y = np.random.randn(200)

# Create scatter plot

plt.scatter(x, y, color='red', s=10)

# Add labels and title

plt.xlabel('X')

plt.ylabel('Y')

plt.title('Scatter Plot of Random X and Y')

# Display graph

plt.show()
```

OUTPUT :